International Journal of Scientific Research in Science and Technology
Available online at : www.ijsrst.com
Print ISSN: 2395-6011 | Online ISSN: 2395-602X doi : https://doi.org/10.32628/IJSRST5231046

Implementation of Traffic Light Controlling System for a Simple Intersection with VHDL using Quartus II

Pranajit Kumar Das
Computer Science and Engineering Department, Sylhet Agricultural University, Bangladesh

ARTICLEINFO

Article History:
Accepted: 01 July 2023
Published: 14 July 2023

Publication Issue
Volume 10, Issue 4
July-August-2023

Page Number
111-117

ABSTRACT

In the era of urbanization in the 21st century, the number of vehicles on the road has increased drastically to cope with the increased population of the city which directly affects the traffic on the street and more importantly on intersections. Automatic traffic signaling system can plays a vital role to mitigate vehicle congestion in the critical intersections of the busy roads. To ensure hassle free mobility of vehicles in the intersection on the street, the appropriate traffic signal lights are necessary that coordinate traffic in various directions on the streets. A simple intersection on the road is considered for the simulation of traffic signaling system. VHDL code was written in Quartus II to implement traffic light system for specific intersections on the road. The vehicles are always moving on the main road, meaning green light is always '1' until there are vehicles approaching the side road. When the vehicles arrive on the side road which is sensed by sensor, then the traffic controller takes the initiative to schedule the traffic light, from green to red light in the main road and red to green light in the side road. The proposed traffic signaling system works properly, which is verified during simulation. In this paper, simple intersection is considered due to simple design and easy implementation.

Keywords: Traffic light system, Road Intersection, VHDL, Quartus II, State transition diagram, Simulation

I. INTRODUCTION

With the rapid economic development globally, especially in middle income countries, the number of automobiles and cars also increased promptly. There is a challenge of traffic on road due to the inconsistency in the development of roads in the city compared to the increased number of vehicles in a city. Therefore, the traffic jams on the road have increased day by day and are becoming more crucial in intra-city communication. Traffic jams are a block situation on the road, caused by several factors, i.e., developed road infrastructure, number of vehicles in the city, traffic signalling system, etc. Traffic jams

Copyright © 2023 The Author(s): This is an open-access article distributed under the terms of the Creative Commons Attribution **4.0 International License (CC BY-NC 4.0)** which permits unrestricted use, distribution, and reproduction in any medium for non-commercial use provided the original author and source are credited.

result in slower vehicle speed, require more travel time, long vehicle queue on the intersection, waste of fuel energy, air pollution affect human health, verbal and physical confrontation on the road, and accidents [1][2].

Traffic controlling system is about more than maintaining and directing vehicles on the road. Because of unplanned urbanization, roads in the cities became more congested day by day, due to large volume of peoples are on the roads for their daily works. An automatic traffic controlling system is an integral part of an effective transportation system in a city. Here are some reasons traffic control systems are so important.

- **Safety:** A traffic control system ensures safety for pedestrians, passengers, drivers, and finally for vehicles. A roadway system without traffic control may become more dangerous.
- **Smooth movement:** In a road intersection, there must be a traffic control system to direct the vehicle on the road. Without maintaining a predefined order in vehicle direction, there would be chaos at road intersections and around intersection areas.
- **Decrease frequency and severity of road accidents:** The traffic signaling system helps to decrease frequency of accidents at intersection and around intersection areas. Drivers become more aware at intersection areas about traffic flows, so the accident become less impactful.
- **Fuel efficiency:** Traffic control system improves the environmental condition as it reduces the time vehicles spend on the road. Usually, the more time vehicles spend on the road, the more carbon they emit.

Traffic management and control is crucial element in any transportation system for safe and systematic operation. Operational guidelines, rules, laws, and devices for signs, and lights are some of the components of any traffic control system. In a road traffic control system, several individual operators are working consistently to work effectively. An automatic traffic signalling system collects vehicle information from the road intersection, analyses and performs accordingly with the traffic situation in intersection [3]. Traffic controlling system ensures trouble-free vehicle flow on road which minimizes the driving time, number of vehicles comes to stop at a time, reduce air pollution, and decrease accidents. There are three colours in any traffic controlling system, namely, red, green, and yellow (Red-Yellow-Green). Those three-color lights control the traffic flows in road intersections.

Traffic Colour	Meaning
	Red light means to stop.
	Yellow light means to wait.
	Green light implies go.

Figure 1: Colour of traffic light with meaning.

Traffic lights, frequently called traffic signals, are usually situated in pedestrian crossings, road intersections, and other locations on the road to control the traffic flows [4]. Currently, modern traffic lighting systems are successfully installed and used smoothly in various cities around the world. Most traffic signal controlling systems have a fixed time cycle which doesn't consider the number of vehicles approaching the directions. This system only switches the light at fixed time intervals. The limitation of this system is sometimes the vehicles wait at empty junction when the red light is on for a cycle. A lot of researchers made contributions on these issues and various improvement has already been made by using sensing devices (sensor) on the lanes with the traffic lights which mitigates the chances of unnecessary waiting for red signal on empty junctions. More recent traffic signalling systems have applied various machine learning approaches which makes the system more sophisticated and real-time. Machine learning (ML) algorithms enable traffic controllers to store vehicle information and measure future waiting time for the vehicles on the lane, where optimized actions are taken with lowest waiting time.

In this paper, a traffic lighting system for a simple intersection is designed using VHDL and implemented in Quartus II software platform. The outline of the paper is organized as follows: Section 2 presents the literature reviews about traffic signalling systems using VHDL. The problem statement is described in section 3. Furthermore, Section 3 covers a simulation of the proposed traffic light system, and the simulation result and discussion are demonstrated in section 4. Finally, Section 5 provides the conclusion of this paper.

II. LITERATURE REVIEW

An A lot of research work has been conducted on traffic light controlling systems by various renowned researchers worldwide. The aim of those conducted research work was to solve various problems associated with traffic controlling systems, several exclusive developments were made but sometimes fail to work with complex real time situation [5][6]. Traffic light controlling system can be implemented using FPGA (Field Programmable Gate Array), Microcontroller and ASIC (Application Specific Integrated Circuit) design. FPGA exhibits many benefits over microcontrollers are performance and amount of input/output. On the other hand, the ASIC design is more costly than FPGA [7][8]. This in-depth literature reviews section presents some recent research works.

B. Dilip et al [9] implemented a FPGA based low-cost traffic light controlling system using ChipScope Pro and Virtual Input Output. The hardware implementation of their design was executed using Spartan-3E FPGA. It was complex and real-time traffic management system in Kingdom of Bahrain, for pedestrians. The authors in [10] proposed a traffic prediction and signal controlling system using the help of IoT devices for a smart city. They used the OWENN (optimized weight Elman neural network) algorithm for traffic prediction and Intel 80286 microprocessor for traffic signal controlling system. Their proposed system involves 5 (five) steps- first, traffic data is collected for the dataset and the information is extracted. Then the extracted data are fed into OWENN classifier as input to find traffic flow. OWENN algorithm achieves 98.23% accuracy and 96.69% F-score over the existing model. In [11], the implementation and simulation of real time traffic light signalling system using FPGA and micro-controller. FPGA and micro-controller were programmed on the Spartan 3E and Arduino platform, respectively. The system was designed with FPGA connected to an electronic circuit board on Spartan 3E through the external expansion and used an Arduino platform connected to the same electronic circuit board. In simulation, C++ and VHDL language is used for Arduino and FPGA programming, respectively. A few hardware components like sensors, switches, LEDs, and resistors are used to build the electronic circuit to ensure the best performance of the traffic light controlling system.

The authors in [12] designed and implemented a traffic light controlling system based on density using GSM for the 4-way road with eight phases. Three PIC 16F877A microcontrollers are used for logic design control, one specially allocated for each of the three sets objectives. The PIC 16F877A microcontrollers is a CMOS FLASH-based 8-bit microcontroller with RISC architecture that allows sufficient pins for the three sets of objectives. Eight pressure (PR) sensors are used on each side of four roadways. C programming language is used to write the set of instructions for the microcontroller to operate. In literature [13], the authors overcome the limitation of fixed time slot in traffic controlling system. They proposed a traffic controlling system with FPGA and sensors that decreases the waiting time. The scheduling of red, yellow, and green light depends on vehicle density on the road intersection. They implemented and tested the proposed system on hardware using ALTERA Cyclone II- FPGA.

III. PROBLEM STATEMENT

A junction (intersection) of two one-way roads is considered in this study, where one road is main road (R1) and other is side road (R2), both roads intersect each other. One sensor is used at the side roadway to sense the presence of vehicles on the sideroad.

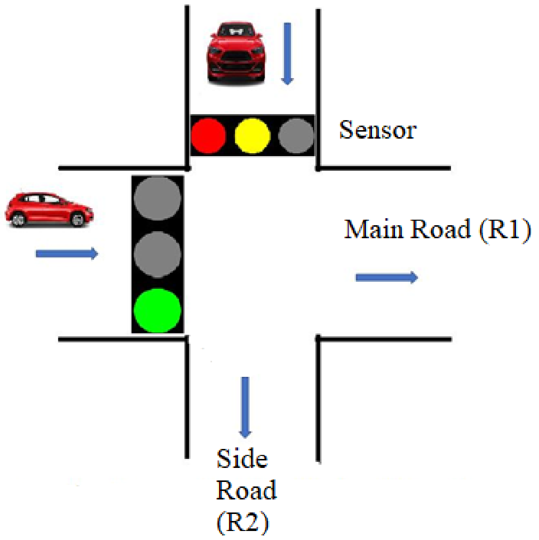


Figure 2: Considered Road junction for designed simulation work.

The traffic light (signal) on the main road (R1) is always green. When any vehicle sensed by sensor on side road (R2) in waiting, the traffic microcontroller first warns the vehicle on main road by yellow light and then red light to stops. The green light on side road turns on and vehicle runs.

IV. IMPLEMENTATION METHODOLOGY

This experimental work was performed on a PC with Windows environment where the OS (operating system) was Windows10, SSD 1T, 16 GB RAM, and NVIDIA GEFORCE RTX 2080 with 16 GB Memory. The Intel Quartus II Web Edition Design Software, Version 13.1 which is a full featured EDA product that was used for the simulation and VHDL programming.

The traffic signaling system for described problem statement is simulated in Quartus II software. Simulation software is used to assess a new design, identify problems in a present system and verify a system under conditions that are difficult to reproduce. In simulation, software permits authority to test, compare, and optimize the results by modelling real world events in a virtual environment.

By default, it is assumed that the vehicles are cars are moving on the main road and there are no cars in the side road. There is a sensor on the side road to sense the presence of the vehicle on the side road. Then the traffic controller transfer light signal for traffic management or to control traffic flow from the side road to main road. For transferring control of traffic lights from main road to side road, there are four states of traffic lighting are defined, which are S0, S1, S2, and S3. The states, their lighting conditions, and state flow are shown in Figure 2.

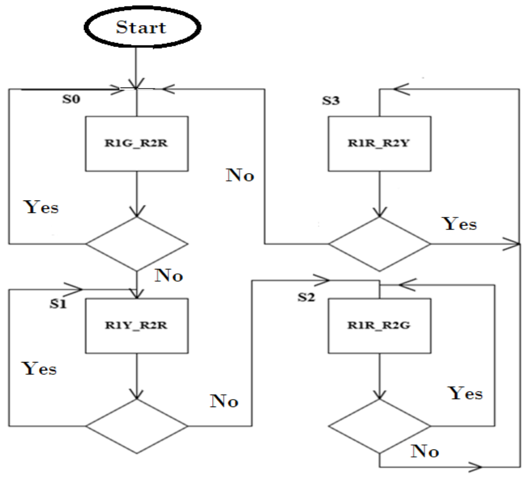


Figure 3: State flow diagram.

At start, the green light is '1' (on) in the main road (R1) and the red light is '1' (on) in the side road (R2), which is called state S0. The traffic light conditions at state S0 are listed below-

In state S0:

Red (R1) =0, Yellow (R1) =0, Green (R1) =1

Red (R2) =1, Yellow (R1) =0, Green (R2) =0

When the sensor placed in the side road (R2) sense vehicle on the side road, the traffic controller warns the vehicles on the main road (R1) by turning on the yellow light. This state is called state S1. Just after 3 seconds, the green light become on in the side road (R2), which is called state S2. The lighting conditions of state S1 and S2 are shown below-

In state S1:

Red (R1) =0, Yellow (R1) =1, Green (R1) =0

Red (R2) =1, Yellow (R1) =0, Green (R2) =0

In state S2:

Red (R1) =1, Yellow (R1) =0, Green (R1) =0

Red (R2) =0, Yellow (R1) =0, Green (R2) =1

When there are no vehicles in the side road, the traffic controller transfer to state S0 through S3. The lighting conditions of S3 are listed below-

In state S3:

Red (R1) =1, Yellow (R1) =0, Green (R1) =0

Red (R2) =0, Yellow (R1) =1, Green (R2) =0

The mnemonic document state diagram of this methodology is shown in Figure 3.

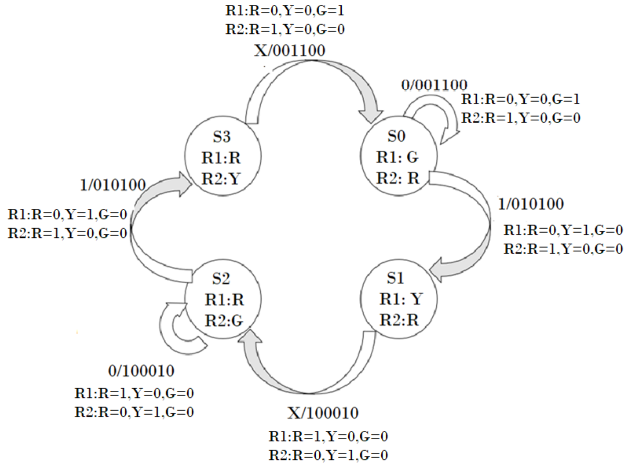
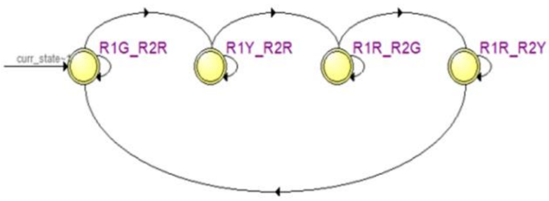


Figure 4 : Mnemonic Document State Diagram

The overall working methodology is shown in Figure 4. According to the traffic flow in the road, the traffic signalling light, green, yellow, and red (G-Y-R) light work in turn. The initial state is R1G_R2R means green on in main road and red on in the side road. The source and destination state are remains same if the sensor have no data. The controller takes 10 seconds when transfer from green to yellow. Yellow to red takes 3 seconds. In the proposed roadway, there are 8 source states and 8 destination states.



(a)

State Table

	Source State	Destination State	Condition
1	R1G_R2R	R1G_R2R	(!sensor_i)
2	R1G_R2R	R1Y_R2R	(sensor_i)
3	R1R_R2G	R1R_R2G	(!delay_10sec)
4	R1R_R2G	R1R_R2Y	(delay_10sec)
5	R1R_R2Y	R1G_R2R	(delay_3sec_f)
6	R1R_R2Y	R1R_R2Y	(!delay_3sec_f)
7	R1Y_R2R	R1R_R2G	(delay_3sec_h)
8	R1Y_R2R	R1Y_R2R	(!delay_3sec...

(b)

Figure 5: State machine viewer with condition of state transition where Fig. 4(a) State of the machine and Fig. 4(b) Conditions for the transition of the state.

V. SOFTWARE SIMULATION RESULT AND DISCUSSION

The proposed traffic signalling system was built using VHDL (Very High-Speed Integrated Circuit Hardware Description Language) on Quartus II. After compilation and debugging, the VHDL code was simulated using Quartus II software platform. The aim of this software was to fully test whether the traffic light controller design goal is achieved or not. After successful compilation of the VHDL code, the register transfer language (RTL) schematics diagram as well as a state machine diagram were automatically generated. RTL viewer file can be viewed as a gate level schematic. The register transfer language (RTL) schematics for the proposed traffic light controlling system are shown in Figure 5.

The automatic generated timing diagram during simulation is shown in Figure 6. From the very beginning, the traffic light condition on the main road (R1) is RYG=001 as shown in the timing diagram.

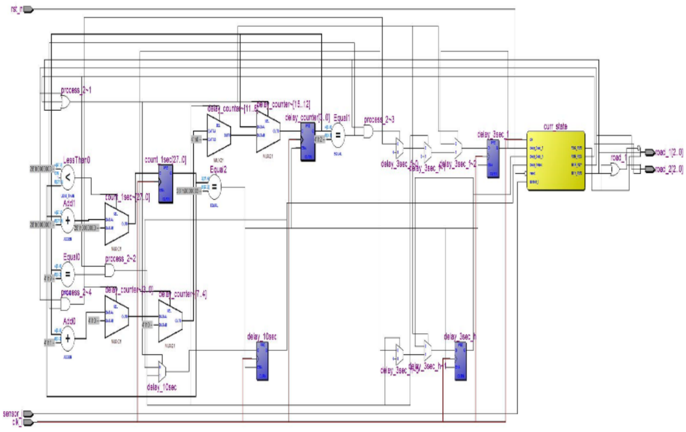


Figure 6: RTL viewer

Whenever the sensor in the side road (R2) sense the presence of the vehicle (sensor is high), the yellow light turn 'high' in main road (R1) means RYG=010 in main road and RYG=100 in the side road. After 3 second delay, the red and green light turn 'high' in the main road (R1) and side road (R2), respectively. The traffic light condition is RYG=100 in main road (R1) and RYG=001 in the side road (R2). The green light becomes 'high' for 10 second in the side road and after defined time it becomes yellow for 3 second before red light being 'high'.

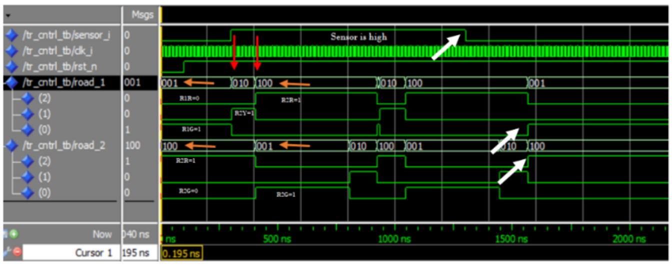


Figure 6 : Timing diagram.

When the sensor data is low (no vehicles in the side road), the green light becomes 'high' in the main road after specified time. The red light becomes 'high' in the side road at the same time. The traffic light conditions are RYG=001 and RYG=100 in the main road and side road, respectively.

VI. CONCLUSION AND FUTURE WORK

A traffic light controlling system for simple intersections was designed and simulated successfully using VHDL in Quartus II software. There is an intersection between the main road and the side road. The reason behind the simple intersection was easy and successful implementation in software simulation. The vehicles are always moving on the main road, meaning green light is always 'on' until there are vehicles approaching the side road. When the vehicles arrive on the side road, then the traffic controller takes the initiative to schedule the traffic light. This case is implemented using VHDL programming language in Quartus II software. State transition diagram with condition, timing simulation diagram, and RTL viewer were generated to test the accuracy and functionality. In this paper, the real-world road intersections are not considered because of simple design and easy implementation. This paper would be helpful for researchers in this field. The future research focuses on vehicle density-based traffic signal controlling systems.

VII. REFERENCES

[1]. Gubbi, J., Buyya, R., Marusic, S., & Palaniswami, M. (2013). Internet of Things (IoT): A vision, architectural elements, and future directions. Future generation computer systems, 29(7), 1645-1660.

[2]. Mehmood, Y., Ahmad, F., Yaqoob, I., Adnane, A., Imran, M., & Guizani, S. (2017). Internet-of-things-based smart cities: Recent advances and challenges. IEEE Communications Magazine, 55(9), 16-24.

[3]. https://www.britannica.com/technology/traffic-control

[4]. Mansuri, F., & Panchal, V. (2016). Four-Way Traffic Light Controller Designing with VHDL. Department of Electronics and Communication, Institute of Technology, Nirma University Ahmedabad, Gujarat, India.

[5]. Nath, S., Pal, C., Sau, S., Mukherjee, S., Roy, A., Guchhait, A., & Kandar, D. (2012, December). Design of an FPGA based intelligence traffic light controller with VHDL. In 2012 International Conference on Radar, Communication and Computing (ICRCC) (pp. 92-97). IEEE.

[6]. Royani, T., Haddadnia, J., & Alipoor, M. (2013). Control of traffic light in isolated intersections using fuzzy neural network and genetic algorithm. International Journal of Computer and Electrical Engineering, 5(1), 142.

[7]. Shanigarapu, L., & Reddy, K. V. (2015). Design and implementation of intelligent traffic light system. International Journal of Computer Science and Mobile Computing (IJCSMC), 4(7), 93-102.

[8]. Bhavana, D., Tej, D. R., Jain, P., Mounika, G., & Mohini, R. (2015). Traffic light controller using FPGA. International Journal of Engineering Research and Applications, 5(4), 165-168.

[9]. Dilip, B., Alekhya, Y., & Bharathi, P. D. (2012). FPGA implementation of an advanced traffic light controller using Verilog HDL. International Journal of Advanced Research in Computer Engineering & Technology (IJARCET), 1(7), 2278-1323.

[10]. Neelakandan, S. B. M. A. T. S. D. V. B. B. I., Berlin, M. A., Tripathi, S., Devi, V. B., Bhardwaj, I., & Arulkumar, N. (2021). IoT-based traffic prediction and traffic signal control system for smart city. Soft Computing, 25(18), 12241-12248.

[11]. Qaddori, S., & Gadawe, N. (2020, September). Real-time traffic light controller system based on FPGA and Arduino. In Proceedings of the 1st International Multi-Disciplinary Conference Theme: Sustainable Development and Smart Planning, IMDC-SDSP 2020, Cyperspace, 28-30 June 2020.

[12]. Nwosu, C., Isiorhovoja, A., Ogbuka, C., & Anyaka, B. (2020). Density based auto traffic light control system with GSM based remote override for enugu metropolis. Journal of applied research and technology, 18(2), 51-61.

[13]. Dabahde, V. V., & Kshirsagar, R. V. (2015). FPGA-based intelligent traffic light controller system design. IJISET-International Journal of Innovative Science, Engineering & Technology, 2(4), 1268-1271.

Cite this article as :

Pranajit Kumar Das, "Implementation of Traffic Light Controlling System for a Simple Intersection with VHDL using Quartus II", International Journal of Scientific Research in Science and Technology (IJSRST), Online ISSN : 2395-602X, Print ISSN : 2395-6011, Volume 10 Issue 4, pp. 111-117, July-August 2023. Available at doi : https://doi.org/10.32628/IJSRST5231046

Journal URL : https://ijsrst.com/IJSRST5231046